HUMAN FEET ARE STRANGE

Humans, along with familiar species such as orangutans, gorillas, lemurs, baboons, and chimpanzees, are primates. Among living primates, modern humans are most closely related to chimpanzees. The two species shared a common ancestor that lived about 6–7 million years ago. One of the traits that distinguish humans from all other primates, including chimpanzees, is the way we walk. Chimpanzees are primarily quadrupedal, which means that they walk on four limbs. Chimpanzees move with their hands turned under so that their knuckles make contact with the ground, which is why they are also described as knuckle-walkers. Humans, on the other hand, are bipedal, meaning that we walk on two legs.

We walk with our feet close together and directly underneath our hips, enabling us to balance on one leg while the other leg swings forward. Chimpanzees can walk on two legs only for short distances. Because their feet are not directly under their hips, they sway side to side to help maintain balance when walking on two legs.

Because humans are the only primate to walk on two legs, bipedality most likely evolved in the lineage that led to humans. When did this human trait first evolve? The “Human Feet Are Strange” activity will help answer that question. Part 1 focuses on observing modern human footprints and what we can infer from them. In Part 2, you will make observations and inferences from a portion of the Laetoli trackway, a trail of footprints that were made in what is now East Africa about 3.6 million years ago! You will compare human feet and chimpanzee feet to the footprints in the Laetoli trackway, to determine whether those footprints were more human-like or chimp-like.

Carefully analyze the two sets of footprints made in class. Consider the following questions when making

your observations:

• Is the big toe pointed in the same direction as the stride?

• How far apart are the left and right feet?

• How long is the stride?

• Where was most of the weight placed on the foot? How can you tell?

Write your notes below.

4. What are the similarities and differences between the two sets of footprints?

Similarities **Differences**

5. If you hadn’t seen the footprints as they were being made, what could you infer about how they were

made or who made them? What observations did you base your inferences on? What couldn’t you infer?

List below the observations and inferences.

Observations **Inferences**

PART 2: LAETOLI TRACKWAY

The Laetoli trackway is a trail of footprints left in soft volcanic ash about 3.6 million years ago that quickly

hardened into a cement-like surface after it was rained on. Unlike the fossils of bones, which are referred to

simply as fossils, fossil footprints are referred to as trace fossils. They provide indirect evidence of past life.

Fossils of Australopithecus afarensis, the same species as the famous “Lucy” fossil, have been found near the

Laetoli footprints. Had A. afarensis made the footprints at Laetoli?

7. Your instructor has shown you an illustration of a small portion of the Laetoli trackway to analyze. Take a

few moments to familiarize yourself with the image. There are two sets of prints in the trackway; one set is

labeled G.1 and the other G.2/3. In each footprint, the outermost lines outline the size of the foot and each

line within represents 1 millimeter of additional depth. The more lines there are encircling an area, the

deeper that part of the footprint was in the ash.

8. What do you observe about the footprints?

a. Is the big toe pointed in the same direction as the stride?

b. Where was most of the weight placed?

9. What can you infer from your observations? For each of the questions below, list the observations and

inferences.

a. How many individuals were walking?

Observation **Inference**

b. What were their relative sizes—that is, how big were they compared with one another?

Observation **Inference**

c. Were they walking together at the same time?

Observation **Inference**

10. Compare the Laetoli footprints to the classroom footprints.

a. What characteristics do they have in common?

b. How are they structurally different from each other?